



Harshitha S 250701234-2025-CSE ▾

H2**Started on** Monday, 5 October 2026, 3:36 PM**State** Finished**Completed on** Monday, 5 October 2026, 3:39 PM**Time taken** 3 mins 17 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8      int dp[n];
9
10     // Read the sequence
11     for (int i = 0; i < n; i++) {
12         scanf("%d", &a[i]);
13         dp[i] = 1;
14     }
15
16     // Find Longest Non-Decreasing Subsequence
17     for (int i = 1; i < n; i++) {
18         for (int j = 0; j < i; j++) {
19
20             // <= because equal values are allowed
21             if (a[j] <= a[i]) {
22                 if (dp[j] + 1 > dp[i]) {
23                     dp[i] = dp[j] + 1;
24                 }
25             }
26         }
27     }
28
29     // Find maximum value in dp[]
30     int answer = dp[0];
31
32     for (int i = 1; i < n; i++) {
33         if (dp[i] > answer) {
34             answer = dp[i];
35         }
36     }
37
38     printf("%d", answer);
39
40     return 0;
41 }
42

```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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